

```
1 #include <systemc.h>
2
3 using namespace sc_core;
4
5 SC_MODULE(RTOS_Scheduler)
6 {
7     sc_event task_event;
8
9     SC_CTOR(RTOS_Scheduler)
10    {
11        SC_THREAD(scheduler);
12    }
13
14    void scheduler()
15    {
16        wait(1, SC_MS);
17
18        cout << sc_time_stamp()
19              << " Task 1 started (Priority 3)" << endl;
20
21        wait(2, SC_MS);
22
23        cout << sc_time_stamp()
24              << " Task 1 completed" << endl;
25
26        wait(1, SC_MS);
27
28        cout << sc_time_stamp()
29              << " Task 2 started (Priority 1)" << endl;
30
31        wait(2, SC_MS);
32
33        cout << sc_time_stamp()
34              << " Task 2 completed" << endl;
35
36        wait(1, SC_MS);
37
38        cout << sc_time_stamp()
39              << " Task 3 started (Priority 2)" << endl;
40
41        wait(2, SC_MS);
42
43        cout << sc_time_stamp()
44              << " Task 3 completed" << endl;
45    }
46 };
47
48 int sc_main(int argc, char* argv[])
49 {
50     RTOS_Scheduler scheduler("RTOS_Scheduler");
51
52     sc_start(15, SC_MS);
53
54     return 0;
55 }
```

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```
[2026-08-23 05:03:45 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim  
Compile done. Starting run...
```

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
```

```
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```
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```

```
1 ms Task 1 started (Priority 3)
```

```
3 ms Task 1 completed
```

```
4 ms Task 2 started (Priority 1)
```

```
6 ms Task 2 completed
```

```
7 ms Task 3 started (Priority 2)
```

```
9 ms Task 3 completed
```

```
Done
```